Learning-Augmented Shoreline Search with Angular Prediction

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Abstract

The Shoreline Search problem requires an agent to locate an infinite line at an unknown distance in the plane. This well-studied problem is optimally solved using logarithmic spirals. This paper takes place in the field of Learning-Augmented Algorithms; the aim is to improve the precedent optimal solution using a shoreline angular prediction about the infinite line. Using this prediction, the problem can be reduced to the interpolation between the linear (1D) and shoreline (2D) problems. We demonstrate that direct Cartesian interpolation fails due to phase desynchronization, and we propose a robust model based on a parametric elliptic spiral. Finally, we provide an analytical derivation of the competitive ratio $c(\eta)$ as a function of the shoreline angular prediction error, establishing the first formal upper bounds for consistency and robustness.

1 Introduction

Algorithmic search theory studies the core strategies that allow an autonomous mobile agent to locate a target whose position is partially or completely unknown. These models provide the mathematical foundation for critical real-world applications, including autonomous robot navigation, search-and-rescue operations in hazardous environments, and data retrieval in unstructured networks.

1.1 Context and Baseline

The mathematical foundation of searching for a target along a single dimension is formalized as the Linear Search Problem (LSP) [5], often referred to as the cow-path problem: an autonomous agent must locate a target positioned on an unbounded line at an unknown distance from the origin. Lacking information about the directional sense, the optimal deterministic strategy requires the agent to execute an alternating sequence of back-and-forth movements with an exponentially expanding search radius.

When the geometric constraints are expanded into a two-dimensional environment, the problem transitions into the Shoreline Search one [7]. In this setting, an agent is placed at an arbitrary origin in a two-dimensional plane with the objective of reaching an infinite straight line (the shoreline) located at a strictly unknown distance. Without any privileged directional information, the agent can no longer rely on simple binary alterations and must instead explore the plane on both axes. Work in the field has shown that this continuous exploration problem can be efficiently addressed using the mathematical family of Elliptic Spirals, which are widely conjectured to provide the optimal worst-case guarantees.

Together, the discrete LSP *(1D)* and the continuous Shoreline Search *(2D)* establish the two classical independent limits of geometric search theory.

1.2 The Learning-Augmented Paradigm

This paper takes place in the emerging field of *Learning-Augmented Algorithms* [11; 9], an architectural framework designed to bridge the gap between classical worst-case analysis and data-driven optimization. In modern applications, search agents are rarely operating in complete blindness; they are frequently equipped with sensors, heuristic predictors, or deep learning oracles that output a *prediction* along with a measure of certainty.

Within this learning-augmented paradigm, the oracle's prediction is leveraged to dynamically improve the runtime efficiency and average-case performance of the search task. Depending on the level of confidence in the received prediction, the algorithm can adapt an internal hyperparameter $\lambda \in [0, 1]$ to balance exploitation of the advice with strategic exploration. However, because machine learning models are fundamentally fallible and can output false prediction, this framework embeds a deterministic algorithmic layer beneath the predictive heuristic. This dual-layer architecture guarantees robust upper bound properties: even if the prediction is completely adversarial, the deterministic layer bounds the agent's efficiency.

1.3 Challenges and Contributions

In the context of the Shoreline Search problem, the *angular shoreline prediction* serves as our untrusted advice [2]. Depending on the confidence in this advice, our objective is to design a learning-augmented algorithm, parameterized by the hyperparameter $\lambda \in [0, 1]$, that dynamically enhances search efficiency when the prediction is accurate, while preserving robust performance guarantees when it fails.

To bridge these paradigms, the proposed model must adhere to two fundamental boundary conditions:

- $\lambda = 0$: The trajectory gracefully reduces to the classical, uninformed 2D shoreline search baseline.
- $\lambda = 1$: The strategy transitions completely into the 1D Linear Search Problem (LSP), utilizing the predicted angle as the primary axis of orientation.

Using this logic, the algorithmic challenge is to create a smooth interpolation (based on λ) between the highly efficient 1D linear search and the robust 2D shoreline search. However, it is hard to interpolate directly on the spatial axes (X and Y coordinates) because the two problems use very different geometries. The 1D search moves back and forth with sharp turns, while the 2D search moves in a continuous, smooth spiral. Mixing these two behaviors directly causes phase desynchronization, creating bad oscillations that ruin the search efficiency.

To overcome this structural incompatibility, we introduce a novel parametric framework. Our primary contributions are summarized as follows:

- We resolve the geometric desynchronization by shifting the interpolation from spatial coordinates to the underlying generative parameters of an elliptic spiral.
- We present a unified analytical framework that synchronizes 1D and 2D search mechanics without topological tearing.
- We provide a comprehensive quantitative evaluation of this system, establishing the first formal consistency and robustness upper bounds for the Shoreline Search problem with angular prediction.

2 State of the Art and Theoretical Foundations

To understand the mechanics of our interpolation, it is essential to characterize the two asymptotic regimes (1D and 2D) that bound our problem and determine how to measure them.

2.1 Learning-Augmented Algorithms

Following the framework by Kumar et al. [9] for rent-or-buy problems like ski rental, we evaluate our algorithm using predictions. In the context of angular prediction for the Shoreline Search, the prediction error η is defined as the absolute angular deviation between the predicted angle θ and the actual one θ':

$$\eta = |\theta - \theta'|$$

The performance of our online algorithm is measured by its competitive ratio, written as a function of the error, $c(\eta)$. Here, $c(0)$ represents the competitive ratio when the prediction error is exactly zero. The evaluation framework relies on two main metrics to bound this ratio:

- **Consistency (β-consistent)**: The competitive ratio achieved when the prediction is perfectly accurate ($\eta = 0$), meaning $c(0) = \beta$.
- **Robustness (γ-robust)**: The competitive ratio in the absolute worst-case scenario, even if the prediction is completely wrong, meaning $c(\eta) \leq \gamma$ for all η.

While these coarse metrics provide a baseline, modern frameworks like the one proposed by Li et al. [10] emphasize *prediction-specific design*, arguing that algorithms should be optimized for the specific value and nature of the prediction received. In our case, the angular shoreline prediction dictates a geometric transformation of the search space.

Our model allows us to smoothly trade off consistency and robustness. While the integration of untrusted advice has been successfully explored for linear search problem [3], according to a recognized repository for this research field, *Algorithms with Predictions* (https://algorithms-with-predictions.github.io/), there are currently no known papers that apply this framework to the Shoreline Search problem and angular predictions.

2.2 One-Dimensional Search (LSP)

Originally introduced by Bellman and optimally solved by Beck and Newman [5], the Linear Search Problem (LSP) involves finding a target hidden on a real line. The searcher starts at the origin (0) but does not know which direction to take or how far away the target is located.

The optimal strategy requires the agent to move back and forth around the origin, turning around and expanding the search radius at each step. This movement follows a geometric progression with a factor of 2, creating a sequence of turns like $1, -2, 4, -8, \ldots$.

Beck and Newman proved that in the worst-case scenario, the total distance traveled by the agent is strictly bounded by 9 times the actual distance D to the target. They also demonstrated that no deterministic search plan can achieve a better worst-case guarantee. Therefore, the optimal competitive ratio for this 1D model is established as $C_{1D} = 9$.

2.3 Two-Dimensional Planar Search

When the direction of the target is completely unknown, the Shoreline Search problem requires continuous exploration in the two-dimensional plane [4; 7]. This problem has been extended to multi-agent scenarios to model robot swarms and collaborative search strategies. Recent breakthroughs have also resolved long-standing open problems regarding different objective functions in planar search [1].

Furthermore, various constrained versions of the problem have been studied, such as searching on a grid where movements are restricted to orthogonal axes, or scenarios where some geometric information (like a partial angular range) is known in advance. These constrained models provide a direct motivation for our approach: if knowing an exact angle or a grid alignment simplifies the search, then a noisy *prediction* of that angle should allow for a flexible, learning-augmented middle ground.

In search theory, it is widely conjectured that logarithmic spirals provide the optimal paths for minimizing the worst-case distance traveled [7]. The standard baseline curve is the symmetric logarithmic spiral given by the polar equation:

$$r(\theta) = e^{\kappa\theta}$$

Assuming this spiral family is optimal, Finch and Zhu [7] computed the exact expansion rate κ. Using a numerical framework, they established the optimal rate at $\kappa_{opt} \approx 0.2124$. This optimal expansion rate yields a worst-case total distance traveled, establishing a 2D competitive ratio of $C_{2D} = 13.81$.

3 Methodology: The Parametric Elliptic Spiral

To simplify the problem, we assume that the predicted angle for the direction of the line is always $\theta = \frac{\pi}{2}$. This means the shoreline associated with the prediction is always a vertical line, which is orthogonal to the x-axis and parallel to the y-axis. Depending on the confidence index $\lambda \in [0, 1]$, the actual orientation of the line can vary around this prediction.

It is easy to generalize this study to an unknown predicted angle by simply rotating the coordinate system to match the prediction.

- **On the X-axis**, the goal is to merge the expanding horizontal movements of both styles. It tries to blend the sharp, back-and-forth jumps of the 1D linear search with the horizontal components of the 2D spiral.
- **On the Y-axis**, the goal is to apply a compression or squash factor based on confidence. As trust in the directional prediction increases, the movement along this perpendicular axis is systematically flattened. When trust is absolute, this orthogonal component becomes exactly zero. This completely removes exploration in the unpredicted direction, collapsing the two-dimensional search space into a single-dimensional line focused entirely along the prediction axis.

3.1 The Failure of Cartesian Interpolation

Let $(X_{1D}(t),\ 0)$ be the one-dimensional trajectory and $(X_{2D}(t), Y_{2D}(t))$ the reference 2D spiral. A direct approach to connect these two problems is to define a new trajectory by linearly interpolating their Cartesian coordinates:

$$P(t, \lambda) = \begin{cases} X(t, \lambda) = \lambda X_{1D}(t) + (1 - \lambda)X_{2D}(t) \\ Y(t, \lambda) = (1 - \lambda)Y_{2D}(t) \end{cases} \tag{1}$$

The main objective of this interpolation is to use the confidence index λ to smoothly transition between the two search domains.

However, we demonstrate that this Cartesian approach is structurally flawed. While the squashing effect on the Y-axis works as intended, combining the two behaviors on the X-axis fails completely due to their fundamentally different geometries. The 1D search proceeds by discrete jumps with sudden changes in direction, whereas the 2D spiral relies on continuous, smooth rotation.

Linearly mixing these two conflicting phase spaces causes severe *phase desynchronization*. Instead of creating a clean and flattened search path, the agent is forced into physically unnatural inflection points. This creates destructive oscillations that severely degrade the competitive ratio. This geometric collapse is visually demonstrated in Figure 1, where the Cartesian interpolation induces sharp, inefficient switchbacks.

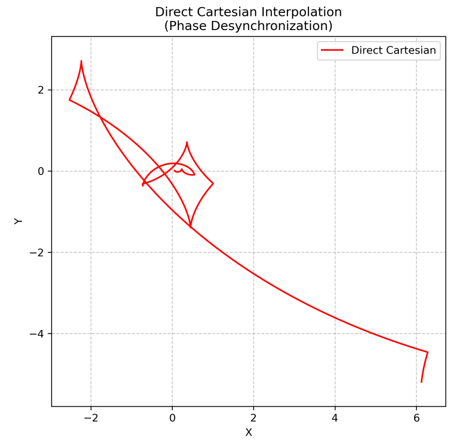

Figure 1: The failure of direct Cartesian interpolation for $\lambda = 0.5$. The trajectory exhibits destructive desynchronization as the phases of the 1D segments and the 2D spiral clash.

Numerous attempts were made to synchronize the X-axis using the raw X functions from both models, but every single trial failed to work.

3.2 Parametric Phase Synchronization

To resolve this desynchronization, we propose interpolating the generative parameters rather than the spatial coordinates.

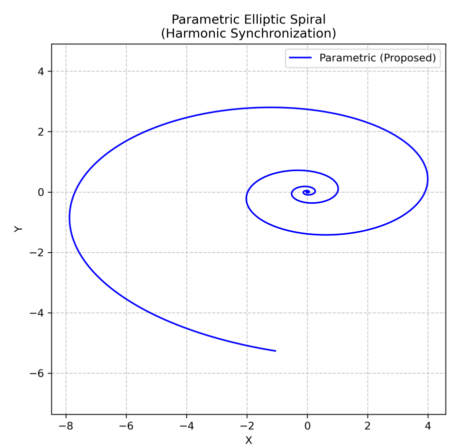

Figure 2: The proposed parametric elliptic spiral for $\lambda = 0.5$. By interpolating generative parameters, the phase is intrinsically synchronized, resulting in a smooth and efficient search path.

For an elliptic spiral to smoothly mimic the back-and-forth movement of the 1D model (where a change of direction translates to a rotation of π), its expansion radius $r(\theta) = e^{a\theta}$ must exactly double at each half-turn:

$$r(\theta + \pi) = 2r(\theta) \implies e^{a(\theta+\pi)} = 2e^{a\theta}$$

By simplifying the exponential terms, we establish the geometric equivalence equation:

$$e^{a_{1\text{D}}\pi} = 2 \implies a_{1\text{D}} = \frac{\ln(2)}{\pi} \approx 0.2206 \tag{2}$$

Let $\lambda \in [0, 1]$ be our confidence index in the angular prediction. We define the interpolated expansion rate:

$$a(\lambda) = \lambda a_{1\text{D}} + (1 - \lambda)\kappa_{opt} \tag{3}$$

The proposed model can then be written analytically in the local frame of the prediction as a parametric elliptic spiral:

$$P(t, \lambda) = \begin{cases} X(t, \lambda) & = e^{a(\lambda)t}\cos(t) \\ Y(t, \lambda) & = (1 - \lambda)e^{a(\lambda)t}\sin(t) \end{cases} \tag{4}$$

For notational simplicity in the subsequent derivations, we treat λ as a fixed parameter and denote $a(\lambda)$ simply as a. Consequently, we will write $P(t)$, $X(t)$ and $Y(t)$.

3.3 Extremum Calculation and Invariant C

To establish the upper bounds of our model, it is necessary to determine the exact moment when the agent narrowly misses the target (*worst-case scenario*).

Derivatives of positions. Assuming the target line is orthogonal to the prediction axis (parallel to the Y-axis), the agent reaches a maximum radial distance on the prediction axis (X) when the derivative of its component vanishes. We will denote t_C a t that satisfy $X'(t) = 0$:

$$\begin{aligned} X'(t) &= \frac{d}{dt}(e^{at}\cos(t)) \\ &= ae^{at}\cos(t) - e^{at}\sin(t) \\ X'(t) &= e^{at}(a\cos(t) - \sin(t)) = 0 \\ &\implies a\cos(t) = \sin(t) \implies \tan(t_C) = a \end{aligned} \tag{5}$$

Symmetrically, assuming the target line is orthogonal to the Y-axis (parallel to the X-axis), the extremum on the orthogonal axis (Y) occurs at angle t_R when $Y'(t) = 0$:

$$\begin{aligned} Y'(t) &= (1 - \lambda)\frac{d}{dt}(e^{at}\sin(t)) \\ &= (1 - \lambda)(ae^{at}\sin(t) + e^{at}\cos(t)) \\ Y'(t) &= (1 - \lambda)e^{at}(a\sin(t) + \cos(t)) = 0 \\ &\implies a\sin(t) = -\cos(t) \implies \tan(t_R) = -\frac{1}{a} \end{aligned} \tag{6}$$

The Angular Intersection Invariant. When the agent misses the target by a distance $\epsilon \to 0$ at a critical point, it must continue turning until it crosses the same infinite target line from behind. Let C be this required phase shift. We demonstrate that whether the agent misses the target on the prediction axis (X) or the orthogonal axis (Y), the catch-up angle C remains exactly the same.

Case 1: Intersection on the Prediction Axis (X-axis). When the agent narrowly misses a line perpendicular to the X-axis at the critical point t_C, the next intersection occurs at $t_{hit} = t_C + C$. The intersection equation is:

$$\begin{aligned} X(t_C + C) &= X(t_C) \\ e^{a(t_C+C)}\cos(t_C + C) &= e^{at_C}\cos(t_C) \\ e^{aC}\cos(t_C + C) &= \cos(t_C) \end{aligned}$$

Using the trigonometric addition formula $\cos(t_C + C) = \cos(t_C)\cos(C) - \sin(t_C)\sin(C)$, we expand the equation:

$$e^{aC}(\cos(t_C)\cos(C) - \sin(t_C)\sin(C)) = \cos(t_C)$$

From our previous calculation for the X-axis extremum, we know the raw relation at t_C satisfies $a\cos(t_C) - \sin(t_C) = 0$, meaning $\sin(t_C) = a\cos(t_C)$. Substituting this directly into the equation yields:

$$\begin{aligned} e^{aC}(\cos(t_C)\cos(C) - a\cos(t_C)\sin(C)) &= \cos(t_C) \\ e^{aC}\cos(t_C)(\cos(C) - a\sin(C)) &= \cos(t_C) \\ e^{aC}(\cos(C) - a\sin(C)) &= 1 \end{aligned} \tag{7}$$

Case 2: Intersection on the Orthogonal Axis (Y-axis). Similarly, if the agent misses a line perpendicular to the Y-axis at the critical point t_R, the next intersection happens at $t_{hit} = t_R + C$. The intersection equation is:

$$\begin{aligned} Y(t_R + C) &= Y(t_R) \\ (1 - \lambda)e^{a(t_R+C)}\sin(t_R + C) &= (1 - \lambda)e^{at_R}\sin(t_R) \\ e^{aC}\sin(t_R + C) &= \sin(t_R) \end{aligned}$$

Expanding the left side with the addition formula $\sin(t_R + C) = \sin(t_R)\cos(C) + \cos(t_R)\sin(C)$ gives:

$$e^{aC}(\sin(t_R)\cos(C) + \cos(t_R)\sin(C)) = \sin(t_R)$$

From our previous Y-axis extremum calculation, the raw relation at t_R satisfies $a\sin(t_R) + \cos(t_R) = 0$, which means $\cos(t_R) = -a\sin(t_R)$. Substituting this directly into our equation yields:

$$\begin{aligned} e^{aC}(\sin(t_R)\cos(C) - a\sin(t_R)\sin(C)) &= \sin(t_R) \\ e^{aC}\sin(t_R)(\cos(C) - a\sin(C)) &= \sin(t_R) \\ e^{aC}(\cos(C) - a\sin(C)) &= 1 \end{aligned} \tag{8}$$

Conclusion. Because Equation 7 and Equation 8 are perfectly identical, the required catch-up angle C is completely independent of the starting axis or the target angle. The phase shift C is a strict invariant of our parametric elliptic spiral model.

4 Quantitative Evaluation: Setting the Upper Bounds

The efficiency of the model is evaluated by dividing the total distance traveled (arc length L) by the actual distance to the target at the moment of the worst-case miss.

4.1 Numerical Solutions and Asymptotic Integration

To compute the exact competitive ratios, our model requires solving two distinct numerical problems: finding the root of the catch-up angle equation and calculating the improper arc length integral.

1. Newton-Raphson Method for the Catch-Up Angle. Equation 8 determines the invariant phase shift C. We define our target root function $f(C)$ as:

$$f(C) = e^{aC}(\cos(C) - a\sin(C)) - 1 = 0$$

To apply the Newton-Raphson algorithm, we take the first derivative of $f(C)$ with respect to C. Using the product rule, the derivative simplifies cleanly to:

$$f'(C) = -e^{aC}(1 + a^2)\sin(C)$$

Using this derivative, the iterative update rule is written as:

$$C_{n+1} = C_n - \frac{f(C_n)}{f'(C_n)} = C_n + \frac{e^{aC_n}(\cos(C_n) - a\sin(C_n)) - 1}{e^{aC_n}(1 + a^2)\sin(C_n)}$$

Because C represents the physical angle needed to turn around and catch up with a missed line, we initialize the algorithm with an initial guess of $C_0 = \pi$. Due to the smooth nature of the spiral, the algorithm achieves rapid quadratic convergence, reaching machine precision within a few iterations.

2. Numerical Quadrature for the Elliptic Arc Length. The calculation of the elliptic arc length has no closed-form solution and requires numerical integration:

$$L(t) = \int_{-\infty}^{t} \sqrt{X'(u)^2 + Y'(u)^2}\, du \tag{9}$$

An integrating limit of $-\infty$ cannot be directly evaluated by computer software. To solve this practically, we leverage the exponential decay of our model. Because the expansion rate parameter satisfies $a > 0$, the term e^{au} shrinks toward zero as u moves in the negative direction.

We can safely truncate the lower bound by replacing $-\infty$ with a small finite starting point T (for example, $T = -50$).

4.2 Consistency and Robustness Bounds

For a given confidence prediction λ, we establish the following competitive ratios:

- **Consistency Upper Bound** $C(\lambda)$: Ratio evaluated at the critical prediction point t_C.

$$C(\lambda) = \frac{L(t_C + C)}{X(t_C)} \tag{10}$$

- **Robustness Upper Bound** $R(\lambda)$: Ratio evaluated at the orthogonal critical point t_R.

$$R(\lambda) = \frac{L(t_R + C)}{Y(t_R)} \tag{11}$$

Conf. λ	Rate a	Consistency $C(\lambda)$	Robustness $R(\lambda)$
0.0	0.2124	13.82	13.82
0.2	0.2140	12.87	17.51
0.5	0.2165	11.64	28.31
0.8	0.2190	10.59	71.39
0.95	0.2202	10.12	286.91
1.0	0.2206	9.00	∞

Table 1: Evolution of the upper bounds as a function of the angular confidence prediction λ. The trade-off is strictly convex.

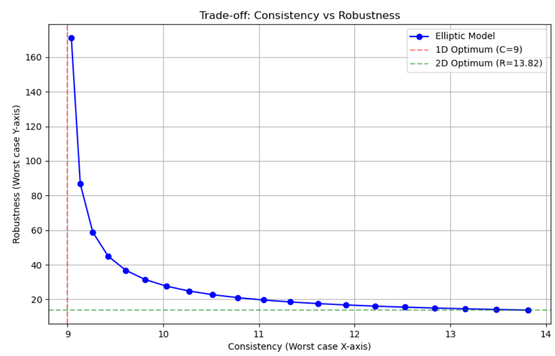

Figure 3: Trade-off between Consistency (Worst-case X-axis) and Robustness (Worst-case Y-axis) as confidence λ varies. The red and green dashed lines indicate the 1D and 2D bounds, respectively.

As illustrated in Figure 3 and detailed in Table 1, the parametric elliptic spiral achieves a smooth, continuous interpolation between the two extremes. When $\lambda = 0$, the algorithm recovers the robust optimal 2D bound of 13.82. As trust λ increases, the consistency improves (approaching the 1D bound of 9.0 on the prediction axis) at the cost of exponentially degrading robustness on the orthogonal axis. This strictly convex trade-off confirms that our generative parameter interpolation successfully mitigates phase desynchronization, allowing for stable, tunable learning-augmented search paths.

4.3 Analysis of the Prediction Error Degradation

A defining characteristic of learning-augmented algorithms is their performance degradation as a function of the prediction error η. While the previous section established the absolute extremes (consistency at $\eta = 0$ and robustness for the worst-case η), we evaluate the competitive ratio $c(\eta)$ continuously as the angular deviation increases.

Let $\eta = |\theta - \theta'|$ be the angular deviation between the shoreline direction angle and the actual one. Let $\vec{n} = (\cos\eta, \sin\eta)$ be the normal of the shoreline direction. To calculate the competitive ratio, we must identify the worst-case distance d for a given trajectory. This corresponds to a shoreline that is perfectly tangent to the agent's path, representing the exact moment the agent misses the shoreline and starts moving away from it.

To evaluate the search performance in an arbitrary direction η, we rotate the frame of reference to align the shoreline's normal vector with the primary axis. In this rotated basis, the agent's depth in the target direction is given by the projection function $f(t, \eta)$:

$$f(t, \eta) = P(t) \cdot \vec{n} = X(t) \cos \eta + Y(t) \sin \eta \tag{12}$$

The critical angle t_{miss} occurs at the peak of this penetration, where the derivative $f'(t, \eta) = 0$:

$$\begin{aligned} f'(t, \eta) &= \frac{d}{dt}\left(e^{at}[\cos t \cos \eta + (1-\lambda) \sin t \sin \eta]\right) \\ &= ae^{at}[\cos t \cos \eta + (1-\lambda) \sin t \sin \eta] \\ &\quad + e^{at}[-\sin t \cos \eta + (1-\lambda) \cos t \sin \eta] \\ &= e^{at}[\cos t(a \cos \eta + (1-\lambda) \sin \eta) \\ &\quad + \sin t(a(1-\lambda) \sin \eta - \cos \eta)] = 0 \end{aligned} \tag{13}$$

Solving for t_{miss} yields:

$$\tan(t_{miss}) = -\frac{a \cos \eta + (1-\lambda) \sin \eta}{(1-\lambda) a \sin \eta - \cos \eta} \tag{14}$$

The optimal distance d to the shoreline is defined as the orthogonal projection of the origin onto the target line. Since the line is tangent to the trajectory at t_{miss}, this distance is exactly the value of the projection function at that moment:

$$d = f(t_{miss}, \eta) \tag{15}$$

Because the phase shift C is a geometric invariant of the model, the agent will intersect the shoreline at $t_{hit} = t_{miss} + C$. The competitive ratio $c(\eta)$ is then calculated as the ratio $\frac{L(t_{hit})}{d}$.

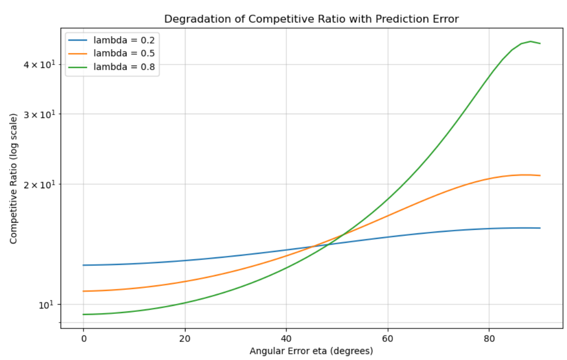

Figure 4: The degradation of the competitive ratio $c(\eta)$ as the angular prediction error η increases from $0°$ to $90°$. Note the logarithmic scale on the Y-axis for higher λ values.

Figure 4 illustrates this degradation. For a conservative confidence parameter like $\lambda = 0.2$, the ratio degrades slowly and remains tightly bounded (maxing out at the robustness bound of 17.51). Conversely, an aggressive parameter like $\lambda = 0.8$ offers superior consistency (10.59) but degrades exponentially as η diverges, highlighting the high risk, high reward nature of heavily trusting the prediction. This analytical framework confirms that our model provides a continuous and predictable response to prediction inaccuracies.

4.4 Utility of the $a(\lambda)$ Interpolation

A critical question in the design of this parametric model is whether the interpolation of the expansion rate $a(\lambda) = \lambda a_{1D} + (1-\lambda)\kappa_{opt}$ is practically necessary, or if the smooth spatial squashing on the Y-axis using the factor $(1-\lambda)$ is sufficient to bridge the two problems.

The optimal 2D expansion rate is $\kappa_{opt} \approx 0.2124$, while the ideal 1D equivalent is $a_{1D} = \frac{\ln(2)}{\pi} \approx 0.2206$. Because these two values are numerically very close, holding the expansion rate constant at either $a(\lambda) = \kappa_{opt}$ or $a(\lambda) = a_{1D}$ yields virtually identical performance bounds.

As shown by our numerical analysis, the difference in competitive ratios is marginal regardless of whether a is interpolated or held constant. For example, at $\lambda = 0.95$, the interpolated model yields a consistency of 9.041, whereas a constant $a = \kappa_{opt}$ yields 9.046. Visually, the corresponding performance curves are nearly indistinguishable. Therefore, from a strictly practical standpoint, the computational overhead of interpolating a is largely unnecessary; the simple suppression of the Y-axis captures almost the entire benefit of the learning-augmented transition.

However, from a philosophical and mathematical perspective, the interpolation remains valuable. It guarantees perfect structural convergence at the extremes: ensuring that the spiral exactly matches the strict geometric properties of the 1D search when $\lambda = 1$, and fully recovers the uncompromised 2D continuous optimum when $\lambda = 0$. While minor in practice, this exact boundary convergence provides rigorous mathematical closure to the parametric synchronization model.

5 Conclusion and Future Work

In this paper, we introduced the first learning-augmented algorithm for the Shoreline Search problem utilizing angular predictions. By transitioning the interpolation logic from raw Cartesian coordinates to the generative parameters of an elliptic spiral, we resolved the severe phase desynchronization that plagues direct interpolation methods. Our unified analytical framework successfully bounds both consistency and robustness, offering a dynamically tunable search strategy that elegantly bridges the Linear Search Problem (1D) and the classical Shoreline Search (2D). We formally established the catch-up angle invariant and provided a comprehensive sensitivity analysis of the competitive ratio $c(\eta)$ regarding angular prediction errors.

Future Work. Several avenues exist for extending this research. First, while we have established the first upper bounds for this problem, further theoretical work is needed to determine if a formal lower bound can be established. Second, the introduction of a random variable into the search strategy could be explored. In the 1D Linear Search Problem, it is well-known that randomized strategies can significantly improve performance, reducing the competitive ratio from the deterministic 9.0 to approximately 4.59 [8]. However, to our knowledge, randomized search remains largely unexplored in the 2D Shoreline Search domain. Investigating how randomness interacts with angular predictions might offer improved average-case performance or tighter bounds by mitigating the risks of deterministic adversarial target placement. Finally, extending this model to handle multiple simultaneous predictions or more complex geometric constraints could provide even greater adaptability in real-world search environments.

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